Module 8 — sPHENIX Fun4All Framework

Week 8 · Running generator + Geant4 inside the collaboration stack

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8.1 What Fun4All is

Fun4All is sPHENIX's analysis/simulation framework, inherited and evolved from PHENIX. It is a C++ / ROOT-based event loop driven by **Subsystem Reco** modules and **Input Managers**. Everything — generator, Geant4, digitization, reconstruction, analysis — is a plug-in module run by a top-level ROOT macro.

8.2 Accessing Fun4All

- **At SDCC (BNL).** Log in and run the sphenix_setup.sh you saw in the Setup Guide. Everything is on cvmfs.
- **Off-site.** Use the public Singularity image ghcr.io/sphenix-collaboration/singularity:latest. Bind-mount /cvmfs and it behaves identically to SDCC.

```
singularity exec --bind /cvmfs --bind $HOME \
docker://ghcr.io/sphenix-collaboration/singularity:latest bash
source /opt/sphenix/core/bin/sphenix_setup.sh -n new
```

8.3 The canonical tutorial macros

Clone the tutorial repository:

```
git clone https://github.com/sPHENIX-Collaboration/tutorials.git
cd tutorials/MyFirstFun4AllMacro
root -b -q -l Fun4All_G4_sPHENIX.C\(100\)
```

This runs 100 events: Pythia generator → Geant4 with the full sPHENIX geometry → output DSTs. For HIJING input:

```
root -b -q -l 'Fun4All_G4_sPHENIX.C(100,"HIJING_AuAu200_mb.hepmc")'
```

8.4 Anatomy of a Fun4All macro

```
int Fun4All_G4_sPHENIX(int nEvents=10, const string& inputFile="") {
auto se = Fun4AllServer::instance();

// --- 1. Input: HepMC file or built-in Pythia8 ---
if (!inputFile.empty()) {
auto in = new Fun4AllHepMCInputManager("HEPMC");
in->AddFile(inputFile);
se->registerInputManager(in);
} else {
PHPythia8* py = new PHPythia8();
py->set_config_file("phpythia8_pp200_minbias.cfg");
se->registerSubsystem(py);
se->registerInputManager(new Fun4AllDummyInputManager("JADE"));
```

```
}

// --- 2. Geant4 ---
G4Setup(0, "FTFP_BERT", true, true, true, true);

// --- 3. Output DST ---
auto out = new Fun4AllDstOutputManager("DST","G4_sPHENIX.root");
se->registerOutputManager(out);

se->run(nEvents);
se->End();
return 0;
}
```

8.5 Reading DSTs

DST files are ROOT files containing PHG4Hit containers, truth tracks, etc. A quick read:

```
root -l G4_sPHENIX.root
root [0] .ls
root [1] DST->GetListOfBranches()->Print();
root [2] DST->Scan("nEvents:eventNumber");
```

8.6 Exercises

- Run the tutorial macro with 100 Pythia min-bias events and inspect the DST.
- Generate a small HIJING file in Module 5's format, feed it to Fun4All_G4_sPHENIX.C.
- Read back the DST and plot the energy in the EMCal tower container for 50 events.

8.7 Self-assessment

- [ ] I launched a Fun4All run either at SDCC or in the Singularity container.
- [ ] I can point to where G4Setup() and PHPythia8 are defined in the repo.
- [ ] I read back at least one DST and extracted one kinematic distribution.